

AADI ARYA

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Education

McMaster University

Sep 2026 - Apr 2031

Honours Bachelor of Computer Science

Hamilton, Canada

- First-year undergraduate student; expected graduation: 2031
- Relevant Coursework: AP Statistics, AP Computer Science, AP Calculus BC, AP Physics 2, Computer Engineering.

Experience

Associate Software Engineering Intern

Jun 2025 – Sep 2025

Canadian Imperial Bank of Commerce (CIBC)

Toronto, ON

- Contributed to an automated risk assessment tool by writing Python and SQL scripts that categorized credit applications, supporting a pilot that cut manual loan review time by an estimated 60% for the team.
- Assisted a senior engineer in testing and refining a BERT based classifier used to flag risk indicators in financial documents, running validation checks against staging data under their guidance.
- Built a front end component of a real time monitoring dashboard in React, working alongside the team's Jenkins based deployment and unit test process.

STEM Workshop Organizer

Jun 2024 – Present

Peel District School Board

Peel Region, ON

- Designed and delivered a hands-on software development curriculum where students built full applications using **React, Flask, TensorFlow, and AWS**.
- Mentored **100+** students in programming concepts including **Python, JavaScript, and C++** through guided coding projects.
- Delivered project-based workshops on coding, circuit building, and hardware-software systems using **React, Flask, TensorFlow, and AWS**.

Organizations

Jun 2023 – Present

Various

Brampton, Canada

- **Microsoft Reactor Program Participant:** Engaged in developer workshops on cloud computing, AI systems, and modern software engineering practices, learning directly from Microsoft engineers and industry professionals.
- **Google Developer Student Clubs:** Collaborated with student developers to build software projects, attend hands-on technical workshops, and explore emerging technologies in AI, cloud, and full-stack development.
- **DECA: 2x Internationals qualifier.** Won best overall in Ontario for Business Law and Ethics.
- **Speech Club President:** Organized events to help **120** students build public speaking and communication skills.

Personal Projects

QuantAlpha: Algorithmic Trading Simulation Engine | *Python, scikit-learn, pytest, Docker*

- Simulated multi-exchange market microstructure (latency, noise, liquidity events) to generate realistic tick-level order flow.
- Built tick-by-tick arbitrage detection filtered by fee/slippage thresholds, executing paper trades on simulated data.
- Trained **GradientBoostingClassifier** and **Ridge** models on lagged returns, reaching **65–67% held-out accuracy**.
- Validated with a 24-test pytest suite and Docker packaging; documented the paper-trading model's limitations honestly.

NeuroScribe: Synthetic Medical Imaging Segmentation Pipeline | *Python, PyTorch, FastAPI, React, boto3/S3, Docker*

- Generated synthetic 2D scans with lesion-like blobs and matching ground-truth masks (no real patient data).
- Trained a **U-Net** (SmallUNet) with combined BCE and Dice loss, cutting validation loss from 1.43 to 0.34 over 20 epochs.
- Added gradient-based interpretability (saliency and Grad-CAM-style heatmaps) via a FastAPI `/segment` endpoint.
- Built S3-compatible storage (boto3, tested with `moto`) and a pytest suite; documented as a non-clinical prototype.

Skills

Programming: Python (ML, CV, OOP), C++, JavaScript, Next.js, MERN Stack, MySQL, C, Java, Linux, Bash, Arduino.
Technical: Data Structures & Algorithms, Cloud Computing (AWS), Tensorflow, Scikit-Learn, Git, Jira, Agile.
Competitions/Certificates: Euclid/CCC Waterloo Math Contests (top 15%), DECA Internationals, Harvard CS50
Fun Facts: I produce music in my free time, play basketball, and enjoy math!